

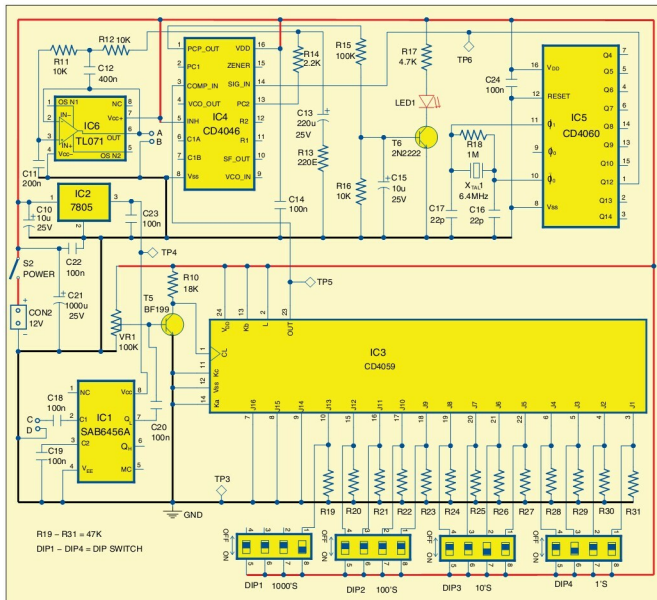


Fig. 2: PLL control section (Unit 2)

CD4060 (IC5) and operational amplifier TL071 (IC6).

This section uses voltage regulator 7805 (IC2) to provide regulated power supply of 5V for the working of IC1. IC3 generates 1.5625kHz reference frequency for the PLL at pin 1, which is fed to frequency comparator (IC4) at pin 14.

A low-level output is taken from the antenna via limiting resistor R9 (shown in Fig. 1), which is coupled to pin 2 of IC1 via capacitor C18. Pin 5 of IC1, the mode pin, is left open to select divide-by-64 mode. Output frequency of the transmitter is divided by 64. Transistor T5 converts the output of IC1 to 12V CMOS

levels. The signal is further divided in programmable divide-by-N counter CD4059 (IC3). Output of CD4059 (pin 23) is connected to the frequency comparator pin 3 of IC4. The PLL comparator (IC4) compares the phase relationship between the reference signal on pin 14 with the input frequency on pin 3.

Depending upon the variance (phase relationship) up or down, a correction voltage is generated on pin 13 of IC4, which is applied to the varactor diode of the VCO to bring it precisely to 'on frequency.' An active low-pass filter (IC6) removes audible 1.5625kHz reference tone from the control voltage.

PLL Transmitter Test Points

Test point	Details
TP0, TP3	0V (GND)
TP1	+12V
TP2	Transmitted frequency
TP4	+5V
TP5	Frequency as set by DIP1-DIP4
TP6	1.5625kHz

Note: All measurements are w.r.t. GND

The reference frequency of the PLL is multiplied by the programmable divider divide rate to give the final frequency. If the divider rate is 1024 (as set by DIP switches DIP1-DIP4